CHEMICAL ENGINEERING 2O04 - Assignment #2

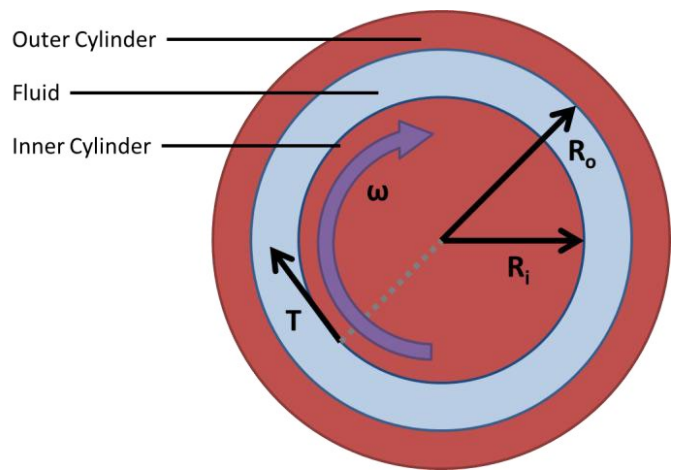
Dr. D.R. Latulippe Due: At 7:00pm on Tue Feb 10th

Notes:

- As per the course outline, full credit is given for completion; thus, your submission will NOT be graded for correctness.
- Your single PDF submission to the Avenue course page should include all your solutions to the questions below.

QUESTION 1

As we discussed in lecture, a concentric cylinder viscometer is often used to measure the viscosity (μ) of Newtonian fluids. As shown in the accompanying figure, the outer cylinder (with radius R_0) remains stationary while the inner cylinder (with radius R_i) rotates at an angular velocity of ω (rev/min) and the torque (T) required to rotate the inner cylinder is measured.



a. Starting from Newton's law of viscosity, derive the correct equation that is used to calculate μ in terms of ω, T, R_0, R_i and ℓ (the height of the fluid in the gap between the two cylinders). Because the radial distance between the cylinders (i.e R_o - R_i) is quite small, you can assume that the velocity profile of the fluid varies linearly with respect to distance between the cylinders.

HINT #1: Torque is the required force multiplied by the moment arm (which is R_i in this case)
HINT #2: Multiply the angular velocity by the circumference of the inner cylinder to get the linear velocity.

Newton's Law of Viscosity $$\tau = \mu \frac{dV}{dy} \quad (1)$$

To solve for viscosity, we need suitable expressions for the shear stress and shear rate.

Let's start with shear stress term: $$\tau = \frac{F}{A} \quad (2)$$

From HINT #1 $$T = F \cdot R_i$$
$$F = \frac{T}{R_i} \quad (3)$$

Substituting (3) into (2): $$\tau = \frac{T}{R_i \cdot A}$$

For a concentric cylinder viscometer, the area (A) corresponds to that for the inner cylinder:

$$A = 2\pi R_i \ell$$

And thus $$\tau = \frac{T}{2\pi {R_i}^2 \ell} \quad (4)$$

Now on to the shear rate term – since we are assuming that the velocity profile of the fluid varies linearly with respect to distance between the cylinders:

$$\frac{dV}{dy} = \frac{V_{outer_cylinder} - V_{inner_cylinder}}{R_o - R_i} = \frac{V_{outer_cylinder}}{R_o - R_i}$$

From HINT #2 $$V_{outer_cylinder} = 2\pi\omega R_i$$

And so $$\frac{dV}{dy} = \frac{2\pi\omega R_i}{R_o - R_i} \quad (5)$$

Substituting (4) and (5) into (1): $$\frac{T}{2\pi {R_i}^2 \ell} = \mu \frac{2\pi\omega R_i}{R_o - R_i}$$

Rearrange to solve for μ: $$\mu = \frac{T(R_o - R_i)}{4\pi^2 \omega {R_i}^3 \ell}$$

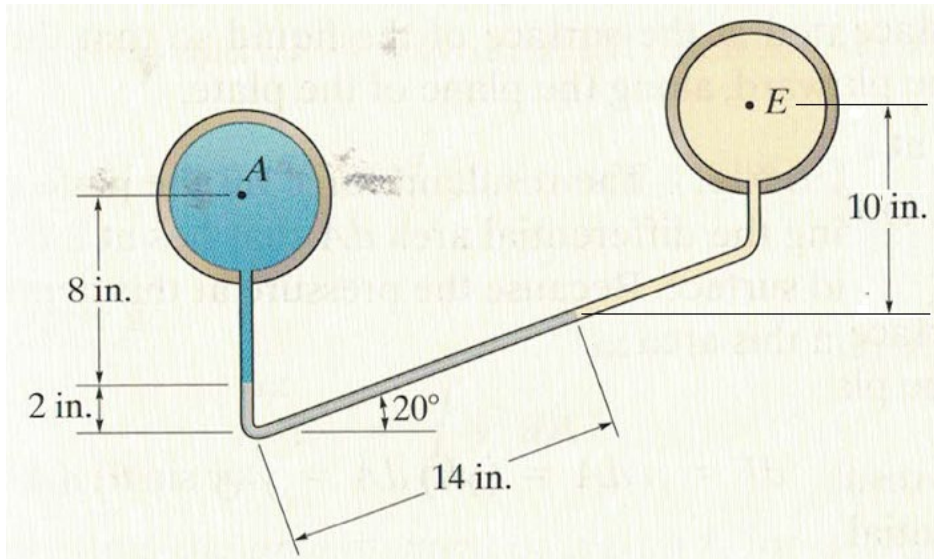
b. For a viscometer with R_i = 12 cm, R_o = 12.15 cm, and ℓ = 40 cm, determine the viscosity (Pa*s) of a Newtonian fluid if a torque of 0.036 N·m is required to rotate the inner cylinder at 500 rev/min.

$$\mu = \frac{0.036 N{\cdot}m(0.1215m - 0.12m)}{4\pi^2\left(500\frac{rev\ 1min}{min 60sec}\right)(0.12m)^3(0.4m)}$$

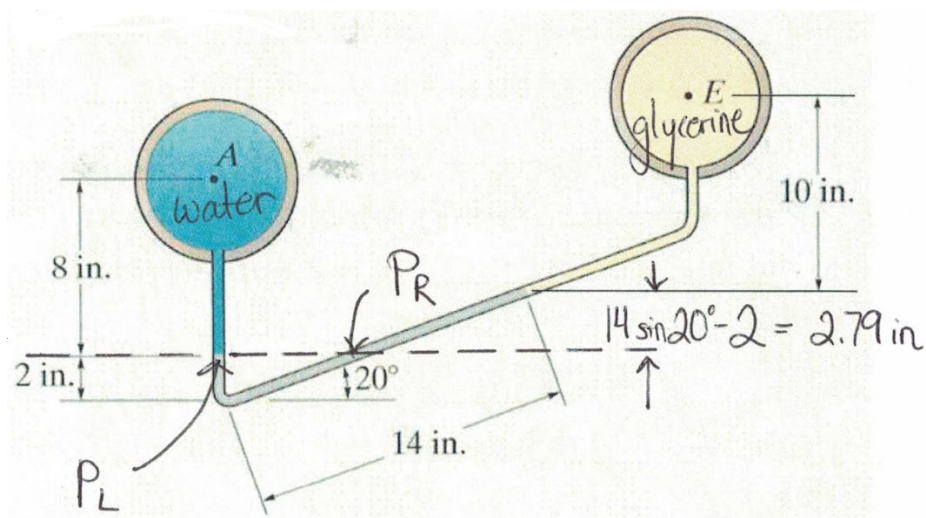
$$\mu = 2.37e^{-4} Pa \cdot s$$

QUESTION 2 (from 2023 Midterm #1)

As shown in the accompanying figure, an inclined-tube manometer filled with mercury is used to measure the pressure difference between point A (at centerline of pipe through which water at 20 °C is flowing) and point E (at centerline of pipe through which glycerine at 20 °C is flowing).



a. Calculate the pressure difference (kPa) between points A and E.



$$P_L = P_R$$

Relating P_A to P_L (according to *"pressure increases linearly with fluid depth")* → $P_L = P_A + \rho_{water} g \left(8in \frac{0.0254m}{1in}\right)$

Relating P_E to P_R (again according to *"pressure increases linearly with fluid depth")* → $P_R = P_E + \rho_{glycerine} g \left(10in \frac{0.0254m}{1in}\right) + \rho_{mercury} g \left(2.79in \frac{0.0254m}{1in}\right)$

Combining three equations:

$$P_A + \rho_{water} g \left(8in \frac{0.0254m}{1in}\right) = P_E + \rho_{glycerine} g \left(10in \frac{0.0254m}{1in}\right) + \rho_{mercury} g \left(2.79in \frac{0.0254m}{1in}\right)$$

Rearrange to solve for $P_A - P_E$:

$$P_A - P_E = \rho_{glycerine} g \left(10in \frac{0.0254m}{1in}\right) + \rho_{mercury} g \left(2.79in \frac{0.0254m}{1in}\right) - \rho_{water} g \left(8in \frac{0.0254m}{1in}\right)$$

At given temperature $\rho_{glycerine} = 1260\ {}^{kg}/_{m^3}$; $\rho_{mercury} = 13550\ {}^{kg}/_{m^3}$; $\rho_{water} = 998\ {}^{kg}/_{m^3}$

And thus $P_A - P_E = 10.6\ kPa$

b. Now consider the scenario where $P_A = P_E$. Write a suitable equation that could be used to solve for the equilibrium position of mercury in the manometer when $P_A = P_E$.

For $P_A = P_E$, the mercury will rise in left leg (by some distance/height h) and fall in the right leg (by same distance h); due to the angle of the manometer, the change in height in the right leg is $h \cdot sin20$

Any point below the dashed line on the figure in part a) could be used to find the equilibrium position – I decided to go right to the bend at the bottom (i.e. where the left leg and right leg connect):

$$P_A + \rho_{water} g(0.203 - h) + \rho_{mercury} g(0.051 + h) = P_E + \rho_{glycerine} g(0.254 + hsin20) + \rho_{mercury} g\big((0.356 - h)sin20\big)$$

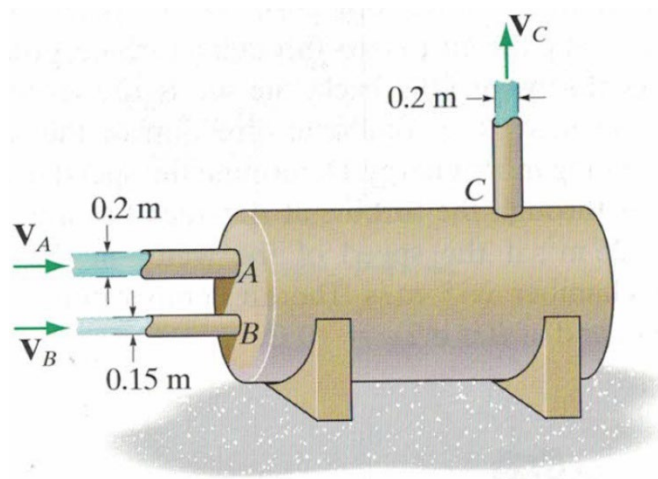
Which can be simplified to:

$$\rho_{water}(0.203 - h) + \rho_{mercury}(0.051 + h) = \rho_{glycerine}(0.254 + hsin20) + \rho_{mercury}\big((0.356 - h)sin20\big)$$

QUESTION 3 (from 2018 Midterm)

Carbon dioxide gas flows into the horizontal cylindrical tank (as shown in figure) at point A at a velocity of 10 m/s; nitrogen gas flows into the same tank at point B at a velocity of 6 m/s. Both gases enter at a gauge pressure of 300 kPa and a temperature of 250 °C. Determine the average velocity of the gas mixture (in m/s) with total density = 1.55 kg/m^3 that leaves the tank at a *steady rate* at point C.

NOTE – you can assume ideal gas properties for both incoming gas streams.



Since it mentions "steady rate",

$$\sum \dot{m}_{in} = \sum \dot{m}_{out}$$

$$\dot{m}_A + \dot{m}_B = \dot{m}_C$$

$$\rho_A V_A A_A + \rho_B V_B A_B = \dot{m}_C$$

$$\left(\frac{P * M_{w,CO2}}{R_u T}\right) V_A A_A + \left(\frac{P * M_{w,N2}}{R_u T}\right) V_B A_B = \dot{m}_C$$

$P_{abs} = P_{gauge} + P_{atm} = 300\ 000 + 101\ 325 = 401\ 325$ Pa

T = 250 + 273.15 = 523.15 K

$$\left(\frac{401325 * 44}{8.314 * 523.15}\right) 10 \left(\frac{\pi}{4}(0.2m)^2\right) + \left(\frac{401325 * 28}{8.314 * 523.15}\right) 6 \left(\frac{\pi}{4}(0.15m)^2\right) = \dot{m}_C$$

$$1275.5\ g/s + 273.9\ g/s = \dot{m}_C$$

$$1594.4\ g/s = \dot{m}_C$$

$$1549.4 \frac{g}{s} = \rho V_C A_C$$

$$V_C = \frac{1549.4 \frac{g}{s} * \frac{1kg}{1000g}}{1.55 \frac{kg}{m^3} * \frac{\pi}{4}(0.2m)^2} = 31.8 \frac{m}{s}$$

QUESTION 4

As shown in the accompanying figure, an inverted conical tank (with radius R and height H) is being used as the feed vessel in a lab-scale membrane filtration setup to evaluate the ability of microfiltration membranes to treat an oily wastewater. A gear pump is used to withdraw the wastewater from the bottom of the tank and send it to a SEPA cross-flow filtration system (made by Sterlitech; https://www.sterlitech.com/sepa-cf-cell.html) at a constant volumetric flow rate (Q). 90% of the flow sent to the filtration system is recycled back to the top of the conical tank; the remaining 10% passes thru the microfiltration membrane and is collected in a separate vessel via the filtrate line.

a. Starting from a general mass balance equation for the conical tank, derive the appropriate equation for the change in liquid level (dh_{tank}/dt).

General mass balance equation $\sum(\dot{m}_{inlet}) - \sum(\dot{m}_{outlet}) = \frac{dm}{dt}$

For the conical tank, there is single inlet and single outlet and so the general equation simplifies to:

$$\dot{m}_{inlet} - \dot{m}_{outlet} = \frac{dm}{dt}$$

Which can be further simplified to: $\rho Q_{inlet} - \rho Q_{outlet} = \frac{d(\rho \forall_{tank})}{dt}$

And since we are dealing with water (an incompressible fluid):

$$\rho Q_{inlet} - \rho Q_{outlet} = \rho \frac{d(\forall_{tank})}{dt}$$

And thus $Q_{inlet} - Q_{outlet} = \frac{d\forall_{tank}}{dt}$

Since the cross-sectional area of the conical tank varies with height, we need to use the "chain rule" for the derivative on the right-hand side:

$$Q_{inlet} - Q_{outlet} = \frac{d\forall_{tank}}{dh}\frac{dh}{dt}$$

As shown in lecture:

- the formula for volume of the conical tank is: $\forall_{tank} = \frac{1}{3}\pi r^2 h$
- by properties of similar triangles: $\frac{r}{h} = \frac{R}{H}$

or $r = \frac{R}{H}h$

so that $\forall_{tank} = \frac{1}{3}\pi\left(\frac{R}{H}h\right)^2 h = \frac{1}{3}\pi\frac{R^2}{H^2}h^3$

and thus $\frac{d\forall_{tank}}{dh} = \frac{1}{3}\pi\frac{R^2}{H^2}3h^2 = \pi\frac{R^2}{H^2}h^2$

Back to our mass balance $Q_{inlet} - Q_{outlet} = \pi\frac{R^2}{H^2}h^2\frac{dh}{dt}$

b. Solve the differential equation from part a) to get the correct expression for the liquid level in the tank as a function of time (i.e. $h_{tank}(t)$); an important condition for you to include is that at the beginning of each filtration test the tank is filled with wastewater to the same height (h_o).

From part a) $$Q_{inlet} - Q_{outlet} = \pi \frac{R^2}{H^2} h^2 \frac{dh}{dt}$$

We are told in the problem that "90% of the flow sent to the filtration system is recycled back to the top of the conical tank" and so the left-hand side simplifies to $Q_{inlet} - Q_{outlet} = -0.1Q$ where Q is the flow rate generated by the gear pump

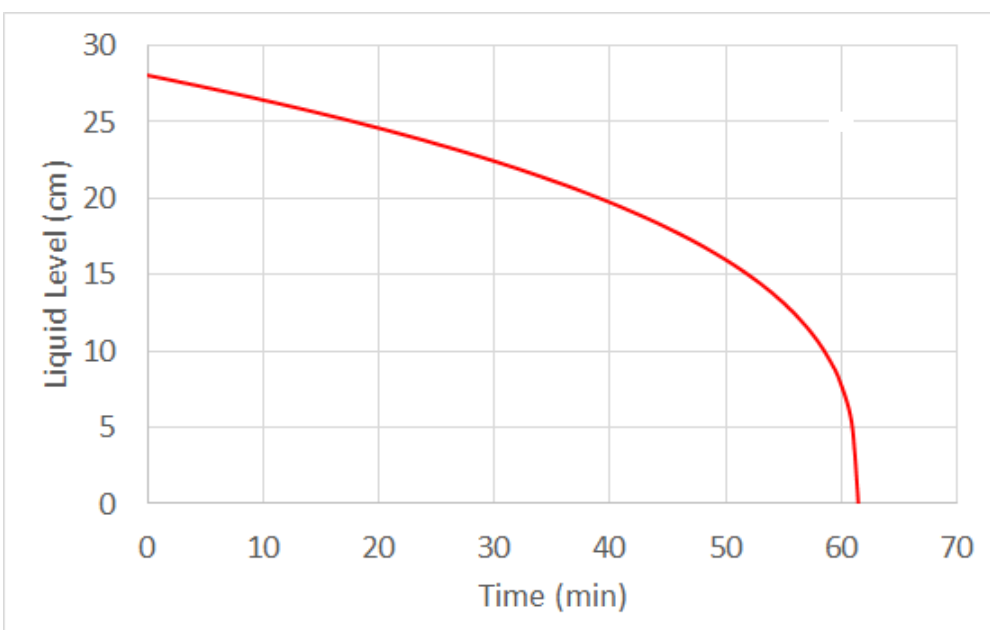
and so $$-0.1Q = \pi \frac{R^2}{H^2} h^2 \frac{dh}{dt}$$

By method of separation of variables $$h^2 dh = -0.1Q \frac{H^2}{\pi R^2} dt$$

$$\int_{h_o}^{h} h^2 dh = -0.1Q \frac{H^2}{\pi R^2} \int_0^t dt$$

$$\frac{1}{3}\left[h^3 - {h_o}^3\right] = -0.1Q \frac{H^2}{\pi R^2} t$$

and so $$h = \left[{h_o}^3 - 0.1Q \frac{3H^2}{\pi R^2} t\right]^{1/3}$$

c. Using Excel, make a nice plot of the wastewater level in the tank (in cm) as a function of time (in min) for the following conditions:

Tank height (H) = 30 cm
Tank radius (R) = 12 cm
Height of wastewater at beginning of filtration test (h_o) = 28 cm
Volumetric flow rate generated by gear pump (Q) = 0.6 L/min
Active area of membrane (A) = 140 cm^2 ←EXTRA INFO THAT IS NOT NEEDED!